REAL GAS PVT CALCULATOR

Thermodynamic Property Prediction Using Cubic Equations of State

Detailed Project Report and User Guide

Topics covered

- Real-gas behaviour and cubic equations of state
- Compressibility, molar volume, density and fugacity
- Saturation pressure, enthalpy and entropy
- Software workflow and step-by-step use
- Worked example using the supplied ethane screenshot
- Interpretation, limitations, validation and future scope

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Abstract

Real gases deviate from ideal-gas behaviour when intermolecular attractions and the finite size of molecules become important, particularly at elevated pressure and near phase-change conditions. This project presents a browser-based Real Gas PVT Calculator that compares four cubic equations of state (EOS): van der Waals (vdW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK), and Peng–Robinson (PR). The supplied program accepts a pure-component gas, temperature and pressure, then estimates compressibility factor, molar volume, density, fugacity coefficient, fugacity, saturation pressure, residual enthalpy and residual entropy. It combines ideal-gas and residual contributions to report total enthalpy and entropy, and plots pressure–volume isotherms for a selected EOS. The report explains the thermodynamic concepts, the program structure, how to operate the calculator, and a worked calculation based on the ethane (C_2H_6) screenshot at 300 K and 50 bar.

Objectives

1. Implement and compare four cubic equations of state under identical conditions.
2. Calculate Z, molar volume and density from the selected EOS roots.
3. Estimate fugacity and saturation pressure when liquid-like and vapour-like roots are available.
4. Separate ideal-gas and residual contributions to enthalpy and entropy.
5. Display the outputs in comparison tables and a pressure–volume graph.
6. Provide a practical user guide and a transparent, step-by-step example calculation.

1. Introduction to PVT Behaviour of Real Gases

PVT describes the relationship among pressure (P), molar volume (V) and temperature (T). For an ideal gas, molecules are treated as point particles with no intermolecular forces, and the equation of state is PV = RT on a molar basis. Real gas molecules occupy finite volume and attract or repel one another. Therefore, the ideal-gas equation becomes less accurate at high pressure and near condensation or critical conditions.

1.1 Compressibility factor

The compressibility factor Z measures departure from ideal-gas behaviour:

$$Z = PV/(RT) \quad \text{and} \quad V = ZRT/P$$

For an ideal gas, Z = 1. If Z is below 1, attractive effects commonly dominate at that state; if Z is above 1, repulsive/excluded-volume effects commonly dominate. Z is not a universal constant: it depends on the gas, temperature and pressure.

1.2 Critical properties and reduced conditions

Each gas is represented by its critical temperature T_c, critical pressure P_c, acentric factor ω and molar mass M. The calculator reports reduced temperature and pressure:

$$T_r = T/T_c \qquad P_r = P/P_c$$

Reduced properties help compare states across different substances. At $T = T_c$ and $P = P_c$, the substance is at its critical point. Above T_c, a conventional liquid–vapour saturation pressure is not defined.

1.3 Why cubic equations of state?

Cubic EOS provide practical corrections to the ideal-gas model while remaining computationally manageable. Their parameters represent attractive intermolecular forces and excluded molecular volume. Solving a cubic EOS in compressibility form may produce one or three real roots. When three physically admissible roots occur, the smallest is interpreted as liquid-like and the largest as vapour-like; the middle root is generally not selected as a stable phase root.

2. Thermodynamic Model: Cubic Equations of State

The calculator uses a generalized cubic equation of state:

$$P = RT/(V - b) - a(T)/[(V + \varepsilon b)(V + \sigma b)]$$

Here P is pressure, T is absolute temperature, V is molar volume, R is the gas constant, a(T) is the temperature-dependent attraction parameter, and b is the co-volume or excluded-volume parameter. The constants ε and σ distinguish the EOS families.

2.1 Models included in the code

EOS	Equation used in the project	Key feature
vdW	$P = RT/(V - b) - a/V^2$	Classic attraction and excluded-volume model.
RK	$P = RT/(V - b) - a/[\sqrt{T}\, V(V + b)]$	Temperature dependence in the attraction term.
SRK	$P = RT/(V - b) - a(T)/[V(V + b)]$	Uses an acentric-factor correction in α.
Peng–Robinson	$P = RT/(V - b) - a(T)/[V(V + b) + b(V - b)]$	Widely used for vapour–liquid equilibrium calculations.

2.2 EOS parameters

The attraction and co-volume parameters are calculated from critical properties:

$$a = \Psi\, \alpha(T_r, \omega)\, R^2 T_c^{\,2}/P_c \qquad b = \Omega\, RT_c/P_c$$

Ω and Ψ are EOS-specific constants. α is a temperature correction. In the supplied code, vdW uses $\alpha = 1$; RK uses $\alpha = T_r^{-1/2}$; SRK and PR use an acentric-factor-dependent expression of the form $\alpha = [1 + m(1 - \sqrt{T_r})]^2$. The value of m differs between SRK and PR.

2.3 Root selection

The program converts the selected EOS to a cubic equation in Z and solves it numerically. It retains roots satisfying $Z > B$, where B is the dimensionless co-volume term used internally. The root selector offers: (i) Auto (stable), which compares fugacity coefficients and chooses the lower-fugacity root when three roots exist; (ii) Vapour-like, which selects the largest root; and (iii) Liquid-like, which selects the smallest root. If only one physical root exists, the program reports a single root.

Important distinction

The 'Root' selector controls which root is used in the property tables. The 'Plot EOS' selector is separate: it chooses which equation of state generates the P–V isotherm. Changing the plotting EOS does not change the selected EOS comparison table.

3. Property Calculations

3.1 Molar volume and density

Once the compressibility root Z is selected, molar volume is calculated from:

$$V = ZRT/P$$

The program uses R = 83.14 cm^3 bar mol^-1 K^-1 for the volumetric calculation, so V is obtained in cm^3/mol when pressure is in bar. Density is then:

$$\rho = M/V$$

With M in g/mol and V in cm³/mol, the code multiplies M/V by 1000 to display kg/m³.

3.2 Fugacity and fugacity coefficient

Fugacity is an effective pressure that accounts for non-ideal behaviour. The fugacity coefficient ϕ measures the departure from ideality:

$$f = \phi P$$

The program evaluates ln ϕ from the chosen EOS and compressibility root, then calculates $\phi = \exp(\ln \phi)$. If $\phi = 1$, fugacity equals pressure and the state behaves ideally in this respect. At a given state, ϕ below one means f is below P.

3.3 Saturation pressure

At liquid–vapour equilibrium, the fugacities of the two phases are equal. At the same pressure this requires $\phi_L = \phi_V$. The code searches for a pressure at which the liquid-like and vapour-like fugacity coefficients become equal. It attempts this calculation only below the critical temperature and when three suitable roots are available. The saturation fugacity is then calculated from $f_{sat} = \phi_{sat} P_{sat}$.

3.4 Enthalpy and entropy

The calculator reports total properties as ideal-gas contributions plus residual (real-gas correction) contributions:

$$H = H_{ig} + H_R \qquad S = S_{ig} + S_R$$

Ideal-gas enthalpy and entropy are obtained by integrating the stored heat-capacity correlations. The residual terms are calculated from the selected EOS and account for non-ideal behaviour. The reference state is an ideal gas at 298.15 K and 1 bar, where H = 0 and S = 0. The code stores approximate Cp correlations intended for roughly 298–1500 K; values outside this range should be treated cautiously.

4. Software Structure and Program Workflow

The calculator is a self-contained browser application. It does not require a Python installation or a server for the supplied version.

Technology	Role in this project
HTML	Defines the gas selector, temperature and pressure inputs, root and plotting selectors, Calculate button, result tables and gra
CSS	Controls layout, colours, spacing, responsive behaviour, input styling, cards and light/dark appearance.
JavaScript	Stores gas data and heat-capacity coefficients; calculates EOS parameters, cubic roots, Z, V, density, fugacity, saturation pre

4.1 Gas database

The supplied code includes methane (CH■), ethane (C■H■), propane (C■H■), butane (C■H■■), nitrogen (N■), oxygen (O■), carbon dioxide (CO■), ammonia (NH■) and water (H■O). For each component, the code stores T_c, P_c, ω and molar mass. A separate Cp data object stores coefficients used for ideal-gas enthalpy and entropy.

4.2 Calculation sequence

1. Read the selected gas, temperature, pressure, root preference and plotting EOS.
2. Load the gas critical properties, acentric factor, molar mass and Cp coefficients.
3. Calculate reduced temperature, reduced pressure, a(T) and b for each EOS.
4. Form and solve the cubic equation in Z; retain physically admissible roots.
5. Select the required root and calculate V, density, ϕ and f.
6. Search for P_{sat} where possible; calculate f_{sat}.
7. Calculate H^R, S^R, H and S.
8. Update the result tables and draw isotherms and the selected pressure line.

4.3 How to open and run the code

1. Copy the complete supplied HTML code into a plain-text editor such as Visual Studio Code or Notepad.
2. Save the file with an .html extension, for example **real_gas_pvt_calculator.html**.
3. Locate the saved file and double-click it, or right-click and choose a modern browser such as Chrome, Edge or Firefox.
4. The calculator should load directly in the browser. An internet connection is not required for the calculations in this version.
5. If you edit the code, save the file and refresh the browser tab to load the updated version.

No build step is needed: the supplied HTML contains the CSS and JavaScript inside the same file. Keep the entire code intact, especially the closing script and HTML tags.

5. Step-by-Step User Guide

5.1 Entering a calculation

1. **Gas:** choose one component from the drop-down list, for example C■H■ (ethane).
2. **Temperature (K):** enter an absolute temperature in kelvin, for example 300.
3. **Pressure (bar):** enter a positive pressure in bar, for example 50.
4. **Root:** choose Auto (stable), Vapour-like or Liquid-like. Use Auto for the program's fugacity-based selection rule.
5. **Plot EOS:** choose vdW, RK, SRK or PR. This controls the isotherm plot only.
6. Press **Calculate**. The information line, comparison tables and P–V graph are refreshed.

5.2 Reading the output

Output	Meaning
EOS / Phase	Equation of state and selected root classification.
Z	Compressibility factor; Z = 1 is the ideal-gas limit.
V (cm³/mol)	Calculated molar volume.
ρ (kg/m³)	Calculated mass density.
P_sat (bar)	Estimated saturation pressure, if the search can find it.
ϕ	Fugacity coefficient from the EOS.
f (bar)	Fugacity at the entered pressure, f = ϕP.
f_sat (bar)	Fugacity at saturation; liquid and vapour values are equal at equilibrium.
H■, S■	Residual enthalpy and entropy contributions.
H, S	Total enthalpy and entropy relative to the stated reference state.

5.3 Reading the graph

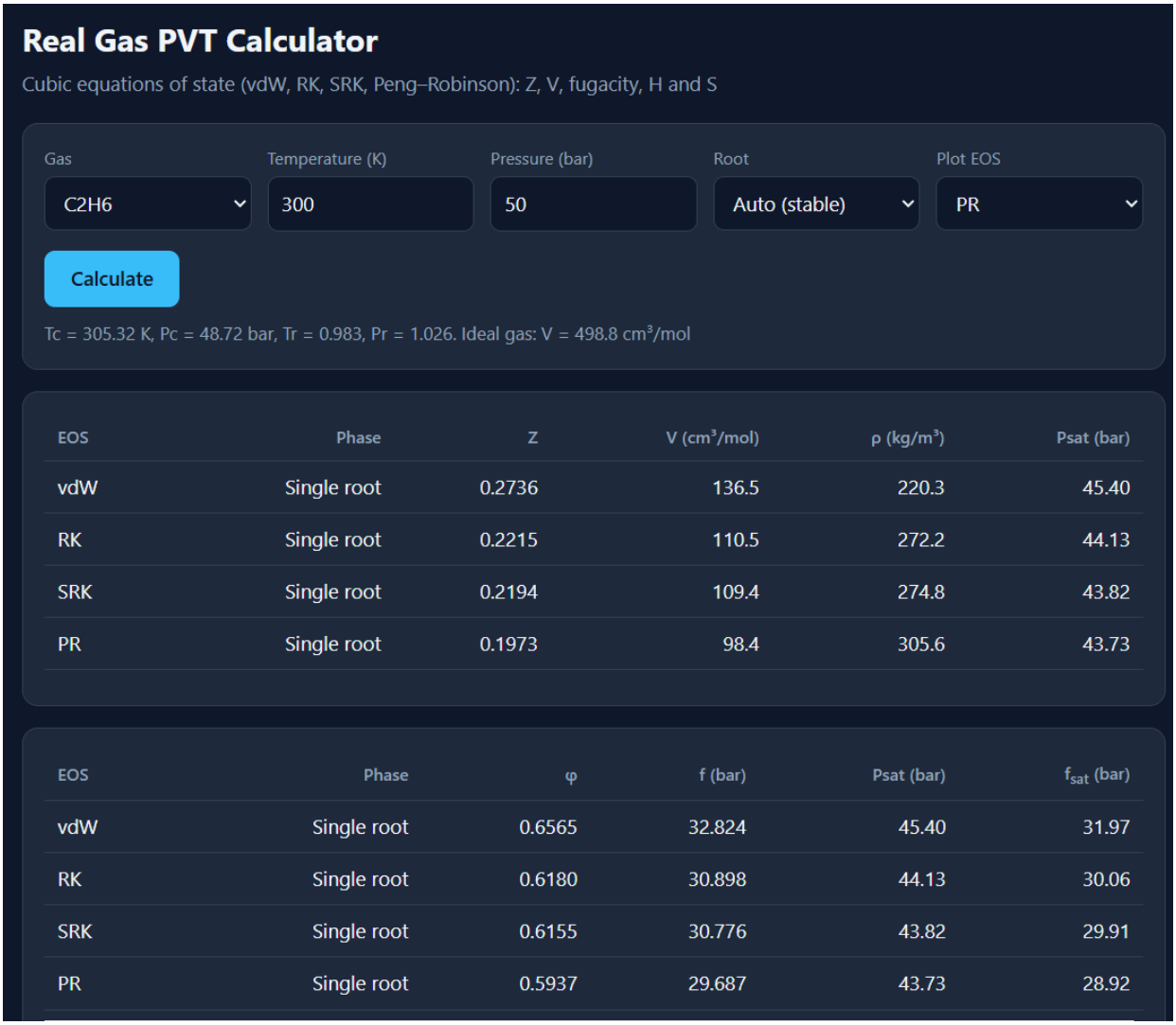
The graph displays pressure against molar volume for the selected plotting EOS. The four thin curves correspond to reduced temperatures T_r = 0.8, 0.9, 1.0 and 1.2. The thicker curve represents the user's entered temperature. The horizontal dashed line represents the entered pressure. At small molar volume the pressure rises sharply because the excluded-volume term becomes important. Below the critical temperature, a cubic EOS may show a non-monotonic region associated with multiple roots.

5.4 Changing conditions

To compare another state, change the gas, temperature, pressure or root preference and press Calculate again. To compare graph shapes from different EOS models, change only Plot EOS. For a meaningful comparison of EOS models, keep gas, T and P fixed while reviewing all four rows in the tables.

6. Application Screenshots

Figure 1 shows the calculator interface with ethane selected at 300 K and 50 bar. The first table compares the four EOS predictions for Z, molar volume, density and saturation pressure. The next table displays fugacity coefficient and fugacity. The visible values are transcribed from the supplied screenshot.



Real Gas PVT Calculator

Cubic equations of state (vdW, RK, SRK, Peng–Robinson): Z, V, fugacity, H and S

Gas	Temperature (K)	Pressure (bar)	Root	Plot EOS
C2H6	300	50	Auto (stable)	PR

Calculate

Tc = 305.32 K, Pc = 48.72 bar, Tr = 0.983, Pr = 1.026. Ideal gas: V = 498.8 cm³/mol

EOS	Phase	Z	V (cm³/mol)	ρ (kg/m³)	Psat (bar)
vdW	Single root	0.2736	136.5	220.3	45.40
RK	Single root	0.2215	110.5	272.2	44.13
SRK	Single root	0.2194	109.4	274.8	43.82
PR	Single root	0.1973	98.4	305.6	43.73

EOS	Phase	φ	f (bar)	Psat (bar)	f_{sat} (bar)
vdW	Single root	0.6565	32.824	45.40	31.97
RK	Single root	0.6180	30.898	44.13	30.06
SRK	Single root	0.6155	30.776	43.82	29.91
PR	Single root	0.5937	29.687	43.73	28.92

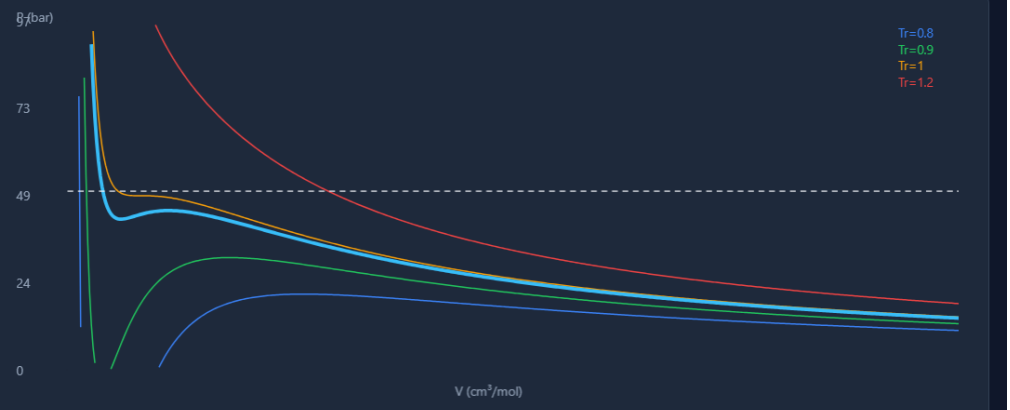
Figure 1. Calculator interface and EOS comparison for ethane at 300 K and 50 bar.

Figure 2 shows the lower part of the same calculation page. It contains the residual and total enthalpy/entropy results, together with the pressure–volume isotherm plot. The selected curve is highlighted, and the dashed horizontal line marks the applied pressure.

Fugacity f = φP, where φ is the fugacity coefficient (ln φ comes from the EOS). At Psat the liquid and vapour fugacities are equal, so one f_{sat} value is shown.

EOS	H^R (J/mol)	S^R (J/mol·K)	H (J/mol)	S (J/mol·K)
vdW	-5899	-16.165	-5801	-48.361
RK	-8435	-24.117	-8337	-56.313
SRK	-9055	-26.148	-8957	-58.344
PR	-9091	-25.970	-8993	-58.166

Reference state: ideal gas at 298.15 K and 1 bar has H = 0 and S = 0. H^R and S^R are residual properties. Ideal-gas Cp data is valid roughly from 298 to 1500 K.



Solid lines are isotherms. The thick line is your T, and the dashed line is your P.

Figure 2. Enthalpy, entropy and pressure-volume isotherms.

7. Worked Example: Ethane at 300 K and 50 bar

For ethane (C2H6) at T = 300 K and P = 50 bar, the Peng-Robinson equation of state is used to demonstrate the property calculations. The critical properties are Tc = 305.32 K, Pc = 48.72 bar, acentric factor = 0.099, and molar mass = 30.07 g/mol.

Step 1 — Calculate reduced temperature and pressure

$$T_r = T/T_c = 300/305.32 = 0.983 \text{ (approximately)}$$
$$P_r = P/P_c = 50/48.72 = 1.026 \text{ (approximately)}$$

These match the information line shown in Figure 1. Since T is slightly below T_c and P is above P_c, the state is close to the critical region; the root structure and model sensitivity deserve attention.

Step 2 — Obtain the PR compressibility factor

The calculator forms the PR cubic equation in Z using the PR constants and the calculated a(T) and b. It solves the cubic and applies the physical-root check. The screenshot reports one physical root for PR, Z = 0.1973. The phase column therefore displays “Single root”.

Step 3 — Calculate molar volume

Use R = 83.14 cm^3 bar mol^-1 K^-1:

$$V = ZRT/P = (0.1973 \times 83.14 \times 300)/50$$
$$V \approx 98.4 \text{ cm}^3/\text{mol}$$

This agrees with the PR volume in Figure 1.

Step 4 — Calculate density

Use M = 30.07 g/mol and the volume just calculated:

$$\rho = (M/V) \times 1000 = (30.07/98.4) \times 1000$$
$$\rho \approx 305.6 \text{ kg/m}^3$$

Step 5 — Calculate fugacity

The calculator evaluates ln ϕ from the PR EOS and selected Z root, then exponentiates it. The screenshot gives ϕ = 0.5937. Therefore:

$$f = \phi P = 0.5937 \times 50 = 29.685 \text{ bar} \approx 29.687 \text{ bar}$$

The small difference is due to rounding of the displayed ϕ. Because ϕ is below 1, fugacity is below the applied pressure for this state.

Step 6 — Saturation pressure and saturation fugacity

The calculator searches below T_c for the pressure at which the liquid-like and vapour-like fugacity coefficients agree. The screenshot reports P_{sat} = 43.73 bar for PR and f_{sat} = 28.92 bar. These are numerical outputs of the code's saturation-pressure search, not independently validated experimental values.

Step 7 — Enthalpy and entropy

From Figure 2, the PR row gives H^R = –9091 J/mol, S^R = –25.970 J/(mol·K), H = –8993 J/mol and S = –58.166 J/(mol·K). Since $H = H^{ig} + H^R$ and $S = S^{ig} + S^R$, the implied ideal-gas contributions from the displayed rounded values are:

$$H^{ig} \approx -8993 - (-9091) = 98 \text{ J/mol}$$
$$S^{ig} \approx -58.166 - (-25.970) = -32.196 \text{ J/(mol·K)}$$

The reference state is ideal gas at 298.15 K and 1 bar with H = 0 and S = 0. The total values are therefore relative to that reference convention.

8. Summary of the Screenshot Results

EOS	Phase	Z	V (cm³/mol)	ρ (kg/m³)	P_sat (bar)
vdW	Single root	0.2736	136.5	220.3	45.40
RK	Single root	0.2215	110.5	272.2	44.13
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EOS	ϕ	f (bar)	P_sat (bar)	f_sat (bar)
vdW	0.6565	32.824	45.40	31.97
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EOS	H■ (J/mol)	S■ (J/mol·K)	H (J/mol)	S (J/mol·K)
vdW	–5899	–16.165	–5801	–48.361
RK	–8435	–24.117	–8337	–56.313
SRK	–9055	–26.148	–8957	–58.344
PR	–9091	–25.970	–8993	–58.166

8.1 Interpretation

All four EOS rows report Z below 1, indicating a substantial departure from ideal-gas behaviour at the selected state. PR gives the smallest Z and molar volume, and the highest density among the four displayed predictions. vdW gives the largest Z and molar volume, and the lowest density. All four fugacity coefficients are below 1, so all four calculated fugacities are below the applied pressure of 50 bar. The saturation-pressure predictions are relatively close, ranging from 43.73 to 45.40 bar.

The enthalpy and entropy values are model estimates. Differences between EOS models arise from their different attraction terms and temperature corrections. The displayed values should be treated as computational predictions rather than validated measurements.

9. Validation Checks and Limitations

9.1 Recommended validation checks

1. **Ideal-gas limit:** at sufficiently low pressure, verify that Z approaches 1 and V approaches RT/P.
2. **Unit consistency:** check that the units of R, P, V and M are consistent in every calculation and conversion.
3. **Root selection:** test states with one and three physical roots and confirm that the selected root matches the requested phase option.
4. **Phase equilibrium:** below T_c, verify that the saturation pressure gives approximately equal liquid and vapour fugacities.
5. **Reference comparison:** compare representative CO■, ethane and other gas cases with a trusted thermodynamic database or published data.
6. **Heat-capacity range:** check the stored Cp correlations against a reliable source over the temperature range in which they are used.

9.2 Limitations of the supplied version

• The code handles pure components only; it does not implement gas-mixture mixing rules.
• Saturation pressure is found by a finite numerical search and may fail if suitable roots or a sign change are not found.
• Cubic EOS accuracy depends on the component and state, and can be weaker near critical conditions or for strongly associating substances.
• Heat-capacity correlations are approximate and are stated to be valid roughly from 298 to 1500 K.
• The screenshot's example values have not been experimentally validated.
• The P–V graph is an illustrative EOS plot; its curves should not be interpreted as experimental data.

9.3 Code-specific caution

The screenshot is an output record for ethane at 300 K and 50 bar, whereas the supplied HTML sets the initial gas selector to CO■ before the user changes it. When reproducing the screenshot, select C■H■ manually. Also, the graph is generated from the EOS selected in the Plot EOS menu and is not a plot of all four EOS simultaneously. Always check the selected gas and input values before recording results.

10. Possible Future Improvements

1. Add mixture calculations using appropriate mixing rules for a_{mix} and b_{mix}.
2. Add a validated thermodynamic property database for comparison and validation.
3. Add downloadable CSV or PDF calculation reports directly from the web application.
4. Add warnings for missing roots, unsuccessful saturation-pressure searches and temperatures outside the Cp correlation range.
5. Add unit selection for pressure, temperature and molar volume, with automatic conversions.
6. Add automated test cases for the ideal-gas limit, critical region, phase equilibrium and known reference states.
7. Add an option to overlay multiple EOS curves on the same P–V plot.

11. Conclusion

The Real Gas PVT Calculator combines four cubic equations of state in a single browser-based tool. It predicts compressibility factor, molar volume, density, fugacity, saturation pressure where possible, and enthalpy and entropy contributions. The tables allow users to compare EOS predictions under the same conditions, while the P–V graph helps visualize how pressure changes with molar volume and temperature. The worked ethane example demonstrates the calculation sequence from reduced properties and compressibility factor through volume, density, fugacity and thermodynamic properties. The next essential step before engineering use is systematic validation against trusted reference data, especially for saturation pressure and heat-capacity-based enthalpy and entropy.